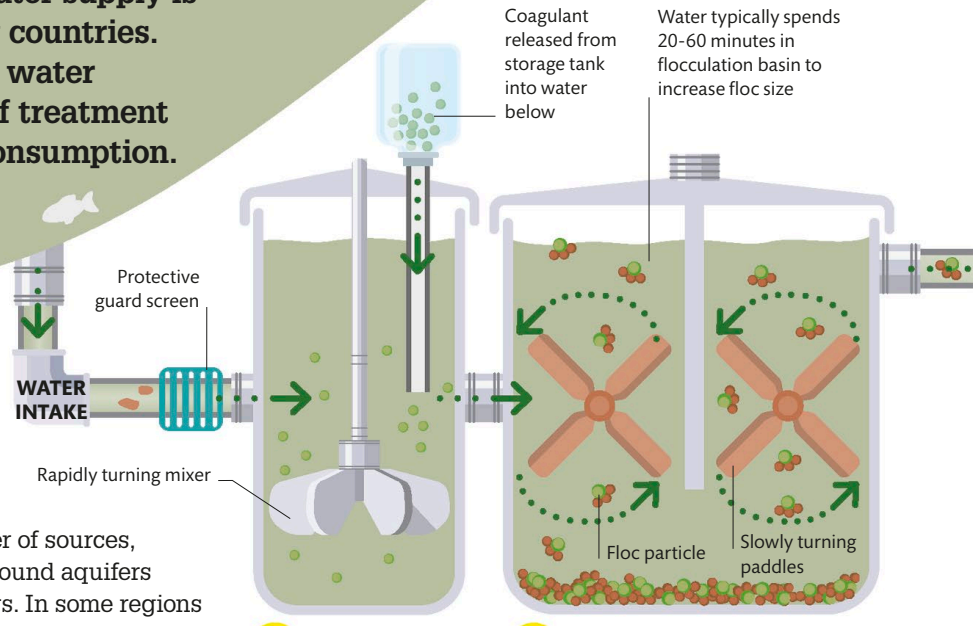
Water supply

A fresh, clean, plentiful water supply is taken for granted in many countries. Before it reaches your tap, water undergoes several types of treatment to make it fit for human consumption.



1 Water intake
Water flows through a series of screens to filter out fish and other waterborne creatures and debris, such as grit, litter, and leaves, preventing them from entering the water-treatment system.

How water is processed

Fresh water is drawn from a number of sources, including lakes, rivers, and underground aquifers (water-bearing rocks), into reservoirs. In some regions lacking plentiful fresh water, desalination plants remove salt from seawater. Whatever the source, the water is purified to remove microorganisms, some of which can cause diseases. Purification also removes harmful chemicals and unwanted odors or tastes before the water is fit for consumption. The water is tested at each stage to monitor its quality.

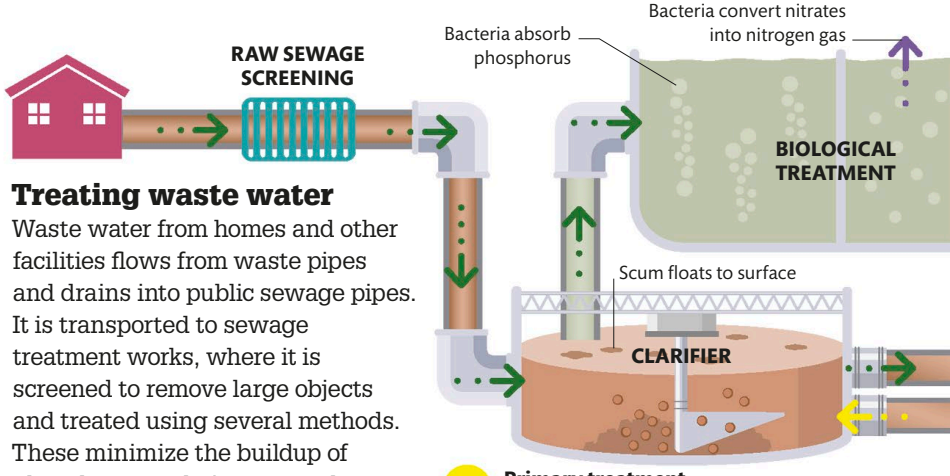
2 Coagulation
The water is mixed rapidly with a coagulant such as ammonium sulphate to help particles suspended in the water collide with each other and clump together.

3 Flocculation
Slowly turning paddles encourage clumped-together particles, called floc, to bind together in larger deposits. These deposits, along with sediment and some bacteria, settle at the bottom of the flocculation basin, while the cleaner water moves to the next stage for further processing.

FLUORIDATION

Fluoride is added to some public water supplies to help add minerals to tooth enamel lost during the process of tooth decay. However, critics claim that overexposure to fluoride in young children can cause "pitting" (small depressions or faults appearing in tooth enamel) and tooth discoloration.





Treating waste water

Waste water from homes and other facilities flows from waste pipes and drains into public sewage pipes. It is transported to sewage treatment works, where it is screened to remove large objects and treated using several methods. These minimize the buildup of phosphorus and nitrogen and remove fats, solid waste particles, and harmful microorganisms.

1 Primary treatment
Solids, such as human waste, settle at the bottom of clarifier tanks and are pumped away; a skimmer removes oil and scum from the surface.